



LUNDS
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Matematikcentrum

Matematik NF

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MATP32

Exercises 1

1. Let $f \in L^1_{loc}(\mathbb{R}^d)$. Show that

$$M_c f(x) \leq M f(x) \leq 2^d M_c f(x) \text{ for a.e. } x \in \mathbb{R}^d.$$

2. Let $f \in L^1(\mathbb{R}^d)$.

- a) Show that Mf is lower semicontinuous.
- b) Conclude that for any $a > 0$, $E_a = \{x \in \mathbb{R}^d : Mf(x) > a\}$ is open.

3. Let $(X, \mathcal{A}, \mu)$ be a measure space and let $1 \leq q < r \leq \infty$. Show that $L^q(X) + L^r(X) \subseteq L^1_{loc}(\mathbb{R}^d)$.

4. Let $f \in L^1(\mathbb{R}^d)$, $f \geq 0$. For $r > 0$ and $x \in \mathbb{R}^d$, let

$$f_r(x) = \frac{1}{|B(r, x)|} \int_{B(r, x)} f(y) dy.$$

- a) Show that $\|f - f_r\|_1 \xrightarrow{r \rightarrow 0} 0$
 - b) Conclude that there exists a strictly decreasing sequence (r_n) , $r_n \xrightarrow{n \rightarrow \infty} 0$ such that $f_{r_n}(x) \rightarrow f(x)$ a.e. on $\mathbb{R}^d$.
5. Let $f \in L^1_{loc}(\mathbb{R}^d)$ and define f_r as in Question 4. For $x \in \mathbb{R}^d$, we define the oscillation of f at x ,

$$\text{osc}(f)(x) = \left| \limsup_{r \rightarrow 0} f_r(x) - \liminf_{r \rightarrow 0} f_r(x) \right|.$$

- a) Prove that $\text{osc}(f) \leq 2Mf$ a.e. on $\mathbb{R}^d$.
- b) Show that for any continuous function h on $\mathbb{R}^d$, $\text{osc}(h)$ is 0 on $\mathbb{R}^d$.
- c) Using the sublinearity of osc , show that for any continuous function h on $\mathbb{R}^d$ and $f \in L^1_{loc}(\mathbb{R}^d)$, $\text{osc}(f) \leq 2M(f - g)$.

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- d) Use the density of $\mathcal{C}_c(\mathbb{R})$ in $L^1(\mathbb{R}^d)$ and the fact that M is weak type $(1,1)$ to conclude that even for $f \in L^1(\mathbb{R})$,

$$\text{osc}(f) = 0 \text{ a.e. on } \mathbb{R}^d.$$

- e) Conclude that the Lebesgue Differentiation Theorem: For $f \in L^1(\mathbb{R})$, $f_r \xrightarrow{r \rightarrow 0} f$ a.e.

The questions will be discussed in the seminar on Thursday, September 10.